

5 Photosynthesis is a process that results in the synthesis of complex organic molecules.

Respiration is a process that involves the breakdown of complex organic molecules.

- (a) The light dependent stage of photosynthesis involves cyclic and non-cyclic photophosphorylation.

Describe **two** differences between cyclic and non-cyclic photophosphorylation.

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..... [2]

- (b) Biochemical processes involving carbon dioxide change the external environment of the aquatic plant, *Elodea canadensis*, and the pond snail, *Lymnaea stagnalis*. Carbon dioxide dissolves in water to form carbonic acid.

A student set up 8 test-tubes. Each test-tube contained 30 cm³ of distilled water containing a pH indicator, bromothymol blue.

Fig. 5.1 shows the experimental set up for these 8 test-tubes.

The test-tubes were left for 12 hours.

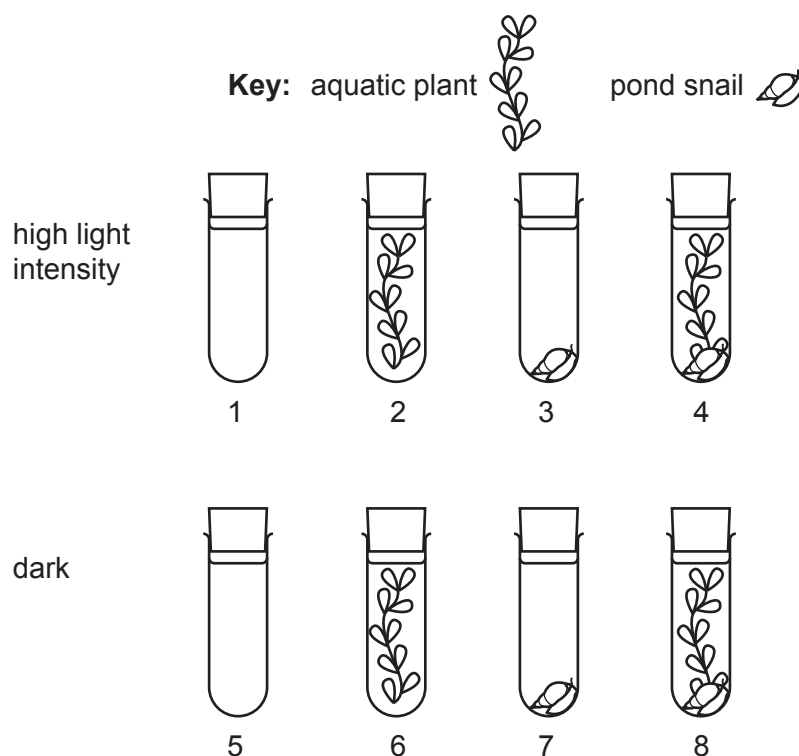


Fig. 5.1

Fig. 5.2 shows how the colour of bromothymol blue changes with pH.

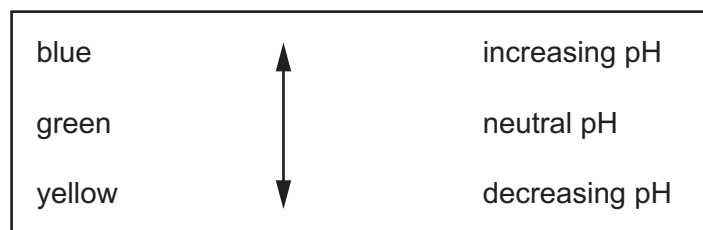


Fig. 5.2

All tubes were green at the start of the experiment.

The results of the experiment are shown in Table 5.1.

Table 5.1

test-tube	colour after 12 hours
1	green
2	blue
3	yellow
4	green
5	green
6	yellow-green
7	yellow
8	yellow

(i) Explain the purpose of test-tube 1 and test-tube 5.

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..... [1]

(ii) Explain the results of test-tubes 2, 3 and 4.

test-tube 2

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test-tube 3

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test-tube 4

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[5]

(iii) Explain the colour change in test-tube 6.

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..... [3]

[Total: 11]